

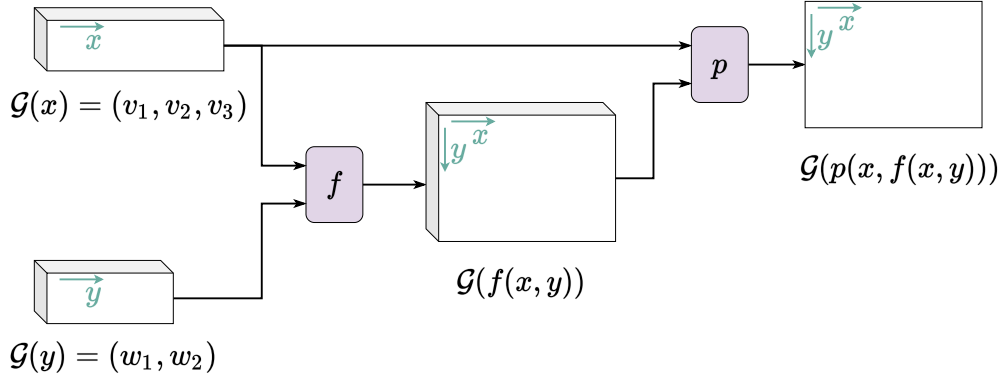


Figure 1: Illustration of Example 3: $\overrightarrow{x}$ and $\overrightarrow{y}$ indicate dimensions associated with the free variables x and y . A tensor representing a term that includes a free variable x will have an axis $\overrightarrow{x}$. One can index $\overrightarrow{x}$ to obtain results calculated using each of the v_1, v_2 or v_3 values of x . In our graphical convention, the depth of the boxes indicates that the tensor can have *feature dimensions* (refer to the end of Example 3).

Example 3. Suppose that $\mathcal{L}$ contains the variables x and y , the function f , the predicate p and the set of domains $\mathcal{D} = \{V, W\}$. Let $\mathbf{D}(x) = V$, $\mathbf{D}(y) = W$, $\mathbf{D}_{\text{in}}(f) = VW$, $\mathbf{D}_{\text{out}}(f) = W$ and $\mathbf{D}(p) = VW$. In what follows, an example of the grounding of $\mathcal{L}$ and $\mathcal{D}$ is shown on the left, and the grounding of some examples of possible terms and atomic formulas is shown on the right.

$$\begin{aligned}
 \mathcal{G}(V) &= \mathbb{R}^+ & \mathcal{G}(f(x, y)) &= \begin{pmatrix} v_1 \cdot w_1 & v_1 \cdot w_2 \\ v_2 \cdot w_1 & v_2 \cdot w_2 \\ v_3 \cdot w_1 & v_3 \cdot w_2 \end{pmatrix} \\
 \mathcal{G}(W) &= \mathbb{R}^- & \mathcal{G}(p(x, f(x, y))) &= \begin{pmatrix} \sigma(v_1 + v_1 \cdot w_1) & \sigma(v_1 + v_1 \cdot w_2) \\ \sigma(v_2 + v_2 \cdot w_1) & \sigma(v_2 + v_2 \cdot w_2) \\ \sigma(v_3 + v_3 \cdot w_1) & \sigma(v_3 + v_3 \cdot w_2) \end{pmatrix} \\
 \mathcal{G}(x) &= \langle v_1, v_2, v_3 \rangle \\
 \mathcal{G}(y) &= \langle w_1, w_2 \rangle \\
 \mathcal{G}(p) &: x, y \mapsto \sigma(x + y) \\
 \mathcal{G}(f) &: x, y \mapsto x \cdot y
 \end{aligned}$$

Notice the dimensions of the results. $\mathcal{G}(f(x, y))$ and $\mathcal{G}(p(x, f(x, y)))$ return $|\mathcal{G}(x)| \times |\mathcal{G}(y)| = 3 \times 2$ values, one for each combination of individuals that occur in the variables. For functions, we can have additional dimensions associated to the output domain. Let us suppose a different grounding such that $\mathcal{G}(\mathbf{D}_{\text{out}}(f)) = \mathbb{R}^m$. Then the dimensions of $\mathcal{G}(f(x, y))$ would have been $|\mathcal{G}(x)| \times |\mathcal{G}(y)| \times m$, where $|\mathcal{G}(x)| \times |\mathcal{G}(y)|$ are the dimensions for indexing the free variables and m are dimensions associated to the output domain of f . Let us call the latter *feature dimensions*, as captioned in Figure 1. Notice that $\mathcal{G}(p(x, f(x, y)))$ will always return a tensor with the exact dimensions $|\mathcal{G}(x)| \times |\mathcal{G}(y)| \times 1$ because, under any grounding, a predicate always returns a value in $[0, 1]$. Therefore, as the "feature dimensions" of predicates is always 1, we choose to "squeeze it" and not to represent it in our graphical convention (see Figure 1, the box output by the predicate has no depth).

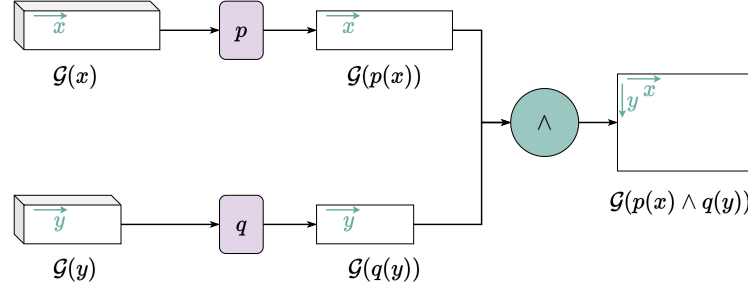


Figure 2: Illustration of an element-wise operator implementing conjunction ($p(x) \wedge q(y)$). We assume that x and y are two different variables. The result has one number in the interval $[0, 1]$ to every combination of individuals from $\mathcal{G}(x)$ and $\mathcal{G}(y)$.

2.2.3. Connectives and Quantifiers

The semantics of the connectives is defined according to the semantics of first-order fuzzy logic [28]. Conjunction ($\wedge$), disjunction ($\vee$), implication ($\rightarrow$) and negation ($\neg$) are associated, respectively, with a t-norm (T), a t-conorm (S), a fuzzy implication (I) and a fuzzy negation (N) operation $\text{FuzzyOp} \in \{T, S, I, N\}$. Definitions of some common fuzzy operators are presented in Appendix B. Let ϕ and ψ be two formulas with free variables $x_1, \dots, x_m$ and $y_1, \dots, y_n$, respectively. Let us assume that the first k variables are common to ϕ and ψ . Recall that $\diamond$ and $\circ$ denote the set of unary and binary connectives, respectively. Formally:

$$\mathcal{G}(\diamond\phi)_{i_1, \dots, i_m} = \text{FuzzyOp}(\diamond)(\mathcal{G}(\phi)_{i_1, \dots, i_m}) \quad (1)$$

$$\mathcal{G}(\phi \circ \psi)_{i_1, \dots, i_{m+n-k}} = \text{FuzzyOp}(\circ)(\mathcal{G}(\phi)_{i_1, \dots, i_k, i_{k+1}, \dots, i_m} \mathcal{G}(\psi)_{i_1, \dots, i_k, i_{m+1}, \dots, i_{m+n-k}}) \quad (2)$$

In (2), $(i_1, \dots, i_k)$ denote the indices of the k common variables, $(i_{k+1}, \dots, i_m)$ denote the indices of the $m - k$ variables appearing only in ϕ , and $(i_{m+1}, \dots, i_{m+n-k})$ denote the indices of the $n - k$ variables appearing only in ψ . Intuitively, $\mathcal{G}(\phi \circ \psi)$ is a tensor whose elements are obtained by applying $\text{FuzzyOp}(\circ)$ element-wise to every combination of individuals from $x_1, \dots, x_m$ and $y_1, \dots, y_n$ (see Figure 2).

The semantics of the quantifiers ($\{\forall, \exists\}$) is defined with the use of aggregation. Let Agg be a symmetric and continuous aggregation operator, $\text{Agg} : \bigcup_{n \in \mathbb{N}} [0, 1]^n \rightarrow [0, 1]$. An analysis of suitable aggregation operators is presented in Appendix Appendix B. For every formula ϕ containing $x_1, \dots, x_n$ free variables, suppose, without loss of generality, that quantification applies to the first h variables. We shall therefore apply Agg to the first h axes of $\mathcal{G}(\phi)$, as follows:

$$\mathcal{G}(Qx_1, \dots, x_h(\phi))_{i_{h+1}, \dots, i_n} = \underset{i_1=1, \dots, |\mathcal{G}(x_1)|}{\text{Agg}(Q)} \underset{i_h=1, \dots, |\mathcal{G}(x_h)|}{\mathcal{G}(\phi)_{i_1, \dots, i_h, i_{h+1}, \dots, i_n}} \quad (3)$$

where $\text{Agg}(Q)$ is the aggregation operator associated with the quantifier Q . Intuitively, we obtain $\mathcal{G}(Qx_1, \dots, x_h(\phi))$ by reducing the dimensions associated with $x_1, \dots, x_h$ using the operator $\text{Agg}(Q)$ (see Figure 3).

Notice that the above *grounded* semantics can assign different meanings to the three formulas:

$$\forall xy(\phi(x, y)) \quad \forall x(\forall y(\phi(x, y))) \quad \forall y(\forall x(\phi(x, y)))$$

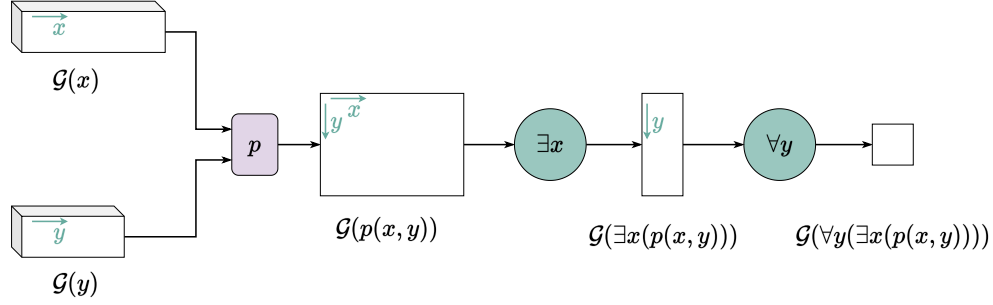


Figure 3: Illustration of an aggregation operation implementing quantification ($\forall y \exists x$) over variables x and y . We assume that x and y have different domains. The result is a single number in the interval $[0,1]$.

The semantics of the three formulas will coincide if the aggregation operator is bi-symmetric.

LTN also allows the following form of quantification, here called diagonal quantification (Diag):

$$\mathcal{G}(Q \text{ Diag}(x_1, \dots, x_h)(\phi))_{i_{h+1}, \dots, i_n} = \text{Agg}(Q) \quad \mathcal{G}(\phi)_{i, \dots, i, i_{h+1}, \dots, i_n} \quad (4)$$

$i=1, \dots, \min_{1 \leq j \leq h} |\mathcal{G}(x_j)|$

$\text{Diag}(x_1, \dots, x_h)$ quantifies over specific tuples such that the i -th tuple contains the i -th instance of each of the variables in the argument of Diag , under the assumption that all variables in the argument are grounded onto sequences with the same number of instances. $\text{Diag}(x_1, \dots, x_h)$ is called diagonal quantification because it quantifies over the diagonal of $\mathcal{G}(\phi)$ along the axes associated with $x_1 \dots x_h$, although in practice only the diagonal is built and not the entire $\mathcal{G}(\phi)$, as shown in Figure 4. For example, given a data set with samples x and target labels y , if looking to write a statement $p(x, y)$ that holds true for each pair of sample and label, one can write $\forall \text{Diag}(x, y) p(x, y)$ given that $|\mathcal{G}(x)| = |\mathcal{G}(y)|$. As another example, given two variables x and y whose groundings contain 10 instances of x and y each, the expression $\forall \text{Diag}(x, y) p(x, y)$ produces 10 results such that the i -th result corresponds to the i -th instances of each grounding. Without Diag , the expression would be evaluated for all 10×10 combinations of the elements in $\mathcal{G}(x)$ and $\mathcal{G}(y)$.⁵ Diag will find much application in the examples and experiments to follow.

2.3. Guarded Quantifiers

In many situations, one may wish to quantify over a set of elements of a domain whose grounding satisfy some condition. In particular, one may wish to express such condition using formulas of the language of the form:

$$\forall y (\exists x : \text{age}(x) > \text{age}(y) (\text{parent}(x, y))) \quad (5)$$

The grounding of such a formula is obtained by aggregating the values of $\text{parent}(x, y)$ only for the instances of x that satisfy the condition $\text{age}(x) > \text{age}(y)$, that is:

⁵Notice how Diag is not simply "syntactic sugar" for creating a new variable pairs_{xy} by stacking pairs of examples from $\mathcal{G}(x)$ and $\mathcal{G}(y)$. If the groundings of x and y have incompatible ranks (for instance, if x denotes images and y denotes their labels), stacking them in a tensor $\mathcal{G}(\text{pairs}_{xy})$ is non-trivial, requiring several reshaping operations.